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11 October 1979
SUPERSEDING
MIL-S-52871A(ME)
10 March 1977

MILITARY SPECIFICATION

SENSOR GROUP, VIBRATION:

DETECTOR DT-546/FSS-9(V) AND PROCESSOR MX-9442/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a vibration signal detector and processor, which combine to form a vibration sensor (VS) designed to detect forcible entry through metal barriers placed over windows and ventilators.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein:

FSC 6350

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-C-45662	- Calibration System Requirements.
MIL-C-52853	- Control Unit, Alarm Set: C-9412/FSS-9(V).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13220E3001	- Processor, Vibration Signal - MX-9442/FSS-9(V).

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TA13220E3013	- Circuit Card Assembly.
TA13220E3051	- Detector, Vibration Signal - DT-546/FSS-9(V).
TA13220E3063	- Circuit Card Assembly.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Vibration Sensor (VS) shall be one processor, MX-9442/FSS-9(V), combined with a minimum of one to a maximum of twenty detectors, DT-546/FSS-9(V). The VS shall be in accordance with Top Assemblies TA13220E3001, TA13220E3051, and as specified herein.

3.1.1 Major components. The VS shall consist of the following major components:

(a) Processor, Vibration Signal TA13220E3001.
(b) Circuit Card Assembly TA13220E3063.
(c) Detector, Vibration Signal TA13220E3051.
(d) Circuit Card Assembly TA13220E3013.

When specified (see 6.2), any of the components listed above shall be furnished separately.

3.1.2 Function. The VS shall detect structurally transmitted vibrations resulting from an attempted forced entry by drilling, cutting, burning, or chipping through protective barriers. The detector(s) shall sense these vibrations and feed an alarm signal to the signal processor when the alarm criteria are met. The detector(s) attached to a uniform structure shall detect vibrations within approximately 3 feet. Up to 20 detectors may be used with a single signal processor to cover large areas or several structures within an area. The signal processor shall draw power from, and supply alarm signals to, a control unit conforming to MIL-C-52853. The processor shall supply power to and accept signals from the detector(s). The processor shall act as a logical "OR" gate between the detector(s) and the control unit.

3.1.3 Drawings. The drawings forming a part of this specification are

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end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the VS or major components (when furnished separately). Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more VS's or major components (when furnished separately), as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce VS's or major components (when furnished separately), that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4. Unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more VS's for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

3.5 Environmental conditions. The VS and major components (when furnished separately), shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.3), performance degradation, or malfunction when tested in accordance with 4.6.2.3.4.

3.5.1 Operational.

(a) Temperature range: -20° F to +150° F.
(b) Humidity up to 95 percent relative.

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3.5.2 Nonoperational and storage.

(a) Temperature range: -30° F to +165° F.
(b) Humidity up to 95 percent relative.
(c) Shock: 20 times the acceleration due to gravity (g's) for 11 milliseconds (ms) duration and bench handling.
(d) Vibration: Transportation conditions.

3.6 Detailed technical requirements.

3.6.1 Terminal board connections.

3.6.1.1 Detector terminal board. The detector terminal board shall be marked with identifying terminal numbers as shown:

Terminal Board 1 (TB1)	Function
1	Vibration detector output signal
2	Vibration detector output signal return
3	Spare
4	Spare
5	Tamper signal
6	Tamper signal
7	Spare
8	Spare
9	Enclosure ground
10	Spare

3.6.1.2 Signal processor terminal boards. The VS signal processor terminal boards shall be marked with identifying terminal numbers as follows:

Terminal Board 1	Function
1	Alarm output
2	Alarm output return
3	Spare
4	Spare
5	Tamper signal
6	Tamper signal
7	+ 20 Volts direct current (Vdc) (from control unit)

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Terminal Board 1	Function
8	20 Vdc return
9	Enclosure ground
10	Spare

On TB1, removable jumpers shall be provided across terminals 2 and 3, 3 and 4, and 4 and 5.

Terminal Board 2 (TB2)	Function
1	Detection input signal
2	Detector input signal return
3	Spare
4	Spare
5	Spare
6	Spare
7	Spare
8	Spare
9	Enclosure ground
10	Spare

3.6.2 Functional characteristics.

3.6.2.1 Power. Power for a VS shall be supplied to the processor from a 20 Vdc regulated power supply in the control unit. Reverse polarity connections to the processor shall not damage the VS. A VS consisting of any number of detectors from one to 20 shall operate as specified herein on regulated voltage between 18 and 22 Vdc with a current drain not exceeding 60 milliamperes (mA).

3.6.2.2 Performance. The VS shall be sensitive to all vibrations in the 10 kilohertz (kHz) to 15 kHz range that result in linear accelerations at the detector enclosure of 0.01 g or greater. The VS shall perform as specified in 3.6.2.2.1, 3.6.2.2.2, and 3.6.2.2.3 under the conditions listed below:

(a) Vibration frequency: 10 kHz to 15 kHz.
(b) Linear acceleration of one detector enclosure: 0.03 g.
(c) Processor sensitivity control: Set to maximum.
(d) The detector gain control: Set to mid-range.
(e) Number of detectors connected to processor: Up to 20.
(f) Cable between processor and each detector: Up to 500 feet of cable, RG 58/U or equal.

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3.6.2.2.1 Alarm criteria. The VS shall be in the alarm state after any detector receives four vibration impulses of 100 ms width within a 15 second period or 4.5 seconds of continuous vibration. Any two or more consecutive impulses occurring within less than 1.5 seconds shall be processed as a single impulse.

3.6.2.2.2 Bandpass characteristics. The combined 3 decibel (dB) bandwidths of the vibration sensor, detector printed circuit board and processor shall be from 10 kHz to 15 kHz. The 30 dB bandwidth shall be from 5 kHz to 20 kHz.

3.6.2.2.3 Sensitivity control range. The processor sensitivity control shall provide a sensitivity adjustment of at least 24 dB.

3.6.2.3 Detailed circuit.

3.6.2.3.1 Gain control range. The detector gain control shall provide a gain adjustment of at least 20 dB.

3.6.2.3.2 Alarm duration. After receipt of alarm criteria, the VS shall be in the alarm state for not less than 0.5 second nor more than 1.5 seconds after the alarm criteria are removed.

3.6.2.3.3 Alarm output. The resistance across the processor alarm output terminals (TB1-1 and TB1-2) shall be as follows:

(a) Non-alarm state: Less than 2 kilohms (2k ohms).
(b) Alarm state: More than 100k ohms.

The alarm output terminals shall be capable of accepting up to 10 mA dc in the non-alarm state and up to 30 Vdc in the alarm state.

3.6.2.3.4 Power interruption. When power to the VS is interrupted or the ground wire is disconnected, the VS shall be in the alarm state.

3.6.2.4 Tamper protection. The resistance between terminals 5 and 6 of the detector and processor TB1 shall represent a tamper alarm condition by exceeding 100k ohms, or a non-tamper condition by being less than 100 ohms. The tamper alarm condition shall be presented whenever the cover is opened from 0.062 to 0.250 inch (flanges engaged) and shall continue until the cover is closed or the tamper switch actuator rod is pulled out. The switch shall reset by closing the cover.

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3.6.2.5 Detector circuit card assembly. The detector circuit card assembly shall be in accordance with applicable drawings TA13220E3063 and have the following characteristics:

(a) Vibration detector circuit output. The vibration detector circuit output shall be 30 millivolts peak to peak (mV P-P) ±3 dB with maximum gain. The output shall be 2 mV ±3 dB with minimum gain. The gain control shall provide a gain adjustment of at least 20 dB.

(b) When the circuit card is installed in the vibration detector enclosure, the tamper switch shall meet the requirements of 3.6.2.4 and the resistance between tamper terminals 5 and 6 shall be as follows:

(1) With actuator rod depressed, less than 100 ohms.
(2) With actuator rod released, more than 100k ohms.
(3) With actuator rod pulled out, less than 100 ohms.

3.6.2.6 Processor circuit card assembly. The processor circuit card assembly shall be in accordance with applicable drawings TA13220E3013 and have the following characteristics:

(a) The 30 dB bandwidth shall be from 5 kHz to 20 kHz.
(b) The sensitivity control shall provide a sensitivity adjustment of not less than 24 dB.
(c) A 11 Vdc ±10 percent pulse of approximately 1.5 seconds shall be produced at test point MV each time a 12 kHz signal burst is applied to terminal TB2-1.
(d) When a 12 kHz continuous signal is applied to terminal TB2-1, a continuous alarm state of zero voltage shall appear at test point INT in 4.5 ±0.5 seconds.
(e) When 4 pulses of 100 ms duration at 12 kHz are applied to TB2-1 within 15 seconds, the alarm state of zero voltage shall appear at test point INT during the fourth pulse and the alarm output shall be greater than 100k ohms.
(f) When 20 Vdc is applied to terminal TB1-1 and signal control switch is in the continuously on position, a 20 V pulse with a duration of 0.5 to 1.5 seconds shall be present at terminal TB1-2 at intervals of approximately 4.5 seconds.

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(g) When the circuit card assembly is installed in the signal processor enclosure, the tamper switch shall meet the requirements of 3.6.2.4 and the resistance between terminals 5 and 6 shall be as follows:

(1) With actuator rod depressed, less than 100 ohms.
(2) With actuator rod released, more than 100k ohms.
(3) With actuator rod pulled out, less than 100 ohms.

3.7 Electromagnetic susceptibility. The electromagnetic susceptibility characteristics of the VS shall conform to MIL-STD-461, Methods RS03, CS01, CS02 and CS06, as modified in 3.7.1 and 3.7.2.

3.7.1 Radiated. Radiated susceptibility shall be conducted in an electric field of 3 Volts per meter (V/m) for both steady-state and keyed conditions of the EMI field generating equipment; this field applies to frequencies up to 500 megahertz (MHz). Above 500 MHz, the electric field shall be 1 V/m.

3.7.2 Conducted. Conducted interference signals shall be applied to both power and signal lines as follows:

(a) Continuous wave; 30 Hz to 400 kHz.
(b) Interference spike voltage; 10 spikes 5 to 10 microseconds wide at a repetition rate of 7 to 10 pulses per second, on input leads only. Random phasing of the spikes with respect to the power line frequency shall be employed.

3.8 Reliability. The VS shall demonstrate a Mean-Time-Between-Failure (MTBF) of 10,000 hours when tested as specified in 4.6.2.4.

3.9 Identification marking. The VS and major components shall be identified in accordance with MIL-STD-130.

3.10 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.8) for testing of major components furnished separately.

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Item No.	Description	Identification	Quantity
1.	Processor, Vibration Signal; MX-9442/FSS-9(V).	TA13220E3001	1
2.	Detector, Vibration Signal; DT-546/FSS-9(V).	TA13220E3051	1

3.11 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or adjoining parts, except for adjustment, and without selection for fit or performance.

3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9, except soldering. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purpose intended.

3.12.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any test or examination by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-loaned property shall be excluded from the requirements of this paragraph.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4.4 and 6.5).

4.2 Classification of inspections. The VS shall be inspected as specified herein. Individual testing of major components installed in the VS is not required. Major components when furnished separately shall be individually inspected as specified herein. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction VS and major components (when furnished separately) shall be examined as specified in 4.6.1. Presence of one or more defect shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

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4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the preproduction VS and major components (when furnished separately), shall be subjected to the following tests.

4.3.2.1 Complete VS. The complete VS shall be subjected to the tests marked "X" in column 1 of Table I. Individual testing of major components of the complete VS is not required. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2.2 Major components furnished separately. Major components (when furnished separately), shall be subjected to the tests marked "X" in Column 1 of Tables II and III. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production VS's will be selected at random by the Government from VS's being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production VS shall not exclude the remaining VS's from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production VS to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production VS and shall be cause for refusal by the Government to continue acceptance of production VS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on VS's previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all VS's accepted prior to completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for tests of the complete VS and major components (when furnished separately), shall be in accordance with MIL-STD-105.

4.5.2 Examination. Each VS and major component (when furnished separately) shall be examined for defects specified in 4.6.1. Presence of one or more defect shall be cause for rejection.

4.5.3 Tests.

4.5.3.1 Individual.

4.5.3.1.1 Complete VS. Each complete VS shall be subjected to the tests marked "X" in Column 3 of Table I. Individual testing of major components of the complete VS is not required. Failure of any test shall be cause for rejection.

4.5.3.1.2 Major components furnished separately. Each major component when furnished separately shall be subjected to the tests marked "X" in Column 2 of Tables II and III. Failure of any test shall be cause for rejection.

4.5.3.2 Samples.

4.5.3.2.1 Complete VS. Complete VS's shall be subjected to the tests marked "X" in Column 4 of Table I. Individual testing of major components of the complete VS is not required. AQL shall be 2.5 percent defective.

4.5.3.2.2 Major components furnished separately. Major components when furnished separately shall be subjected to the tests marked "X" in Column 3 of Tables II and III. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The VS and major components (when furnished separately), shall be examined as specified herein for the following major defects:

101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.
102. Components missing or not as specified.
103. Assembly not correct.

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104. Electromagnetic interference suppression components and network not identical to the approved model.
105. Identification or special marking missing, incomplete or illegible.
106. Interchangeability not as specified.
107. Treatment and painting not as specified.
108. Workmanship including wiring and soldering not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. Prior to testing, each VS shall be subjected to a 72-hour burn-in, during which the signal processor shall be set to maximum gain and the detector shall be set 10 dB below maximum gain. The VS shall be exercised through all its functions not less than nine times during each 24 hours of burn-in. All failures shall be recorded and analyzed for cause of failure. Corrective action shall be taken when appropriate. Copies of these failure reports shall be forwarded to the contracting officer. Units for testing shall be selected from those units having successfully completed the 72-hour burn-in.

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4.6.2.2 Test schedule.

TABLE I. Test schedule for complete vibration sensor.

Preproduction	Initial Production	Quality Conformance		Tests	Requirement paragraph	Test paragraph
		Individual	Sample			
1	2	3	4	5	6	7
				Functional:		
X	X	X	-	(a) Power	3.6.2.1	4.6.2.3.1.1
X	X	X	-	(b) Performance	3.6.2.2	4.6.2.3.1.2
X	X	-	X	(c) Detailed circuit	3.6.2.3	4.6.2.3.1.3
X	X	-	X	(d) Tamper switch tolerance	3.6.2.4	4.6.2.3.1.4
X	X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.3.4.1
X	X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.3.4.2
X	X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.3.4.3
X	X	-	-	Shock	3.5.2(c)	4.6.2.3.4.4
X	X	-	-	Vibration	3.5.2(d)	4.6.2.3.4.5
X	X	-	-	Electromagnetic susceptibility	3.7	4.6.2.3.5
X	X	-	-	Reliability 1/	3.8	4.6.2.4

1/ When specified (see 6.2) preproduction reliability tests can be deleted at time of contract award based upon equipment performance, past history, and contractor past performance.

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TABLE II. Test schedule for detector circuit card assembly.

Preproduction	Quality Conformance		Tests	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional:	3.6.2.5	4.6.2.3.2
X	-	X	Low temperature 1/	3.5.1(a), 3.5.2(a)	4.6.2.3.4.1
X	-	X	High temperature 1/	3.5.1(a), 3.5.2(a)	4.6.2.3.4.2
X	-	X	Humidity 1/	3.5.1(b), 3.5.2(b)	4.6.2.3.4.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.3.4.4
X	-	-	Vibration 1/	3.5.2(d)	4.6.2.3.4.5
X	-	-	Electromagnetic susceptibility 1/	3.7	4.6.2.3.5

1/ Place detector circuit card assembly in a Government-loaned VS or equivalent to simulate normal environment when installed in VS.

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TABLE III. Test schedule for processor circuit card assembly.

Preproduction	Quality Conformance: Individual	Quality Conformance: Sample	Tests	Requirement paragraph	Test paragraph
1	2	3	4	5	6
X	X	-	Functional:	3.6.2.6	4.6.2.3.3
X	-	X	Low temperature 1/	3.5.1(a), 3.5.2(a)	4.6.2.3.4.1
X	-	X	High temperature 1/	3.5.1(a), 3.5.2(a)	4.6.2.3.4.2
X	-	X	Humidity 1/	3.5.1(b), 3.5.2(b)	4.6.2.3.4.3
X	-	-	Shock 1/	3.5.2(c)	4.6.2.3.4.4
X	-	-	Vibration 1/	3.5.2(d)	4.6.2.3.4.5
X	-	-	Electromagnetic susceptibility 1/	3.7	4.6.2.3.5

1/ Place processor circuit card assembly in a Government-loaned VS or equivalent to simulate normal environment when installed in VS.

4.6.2.3 Test methods. Tests shall be conducted in accordance with the test methods in the appendix and as specified herein. Calibration of test equipment shall be in accordance with MIL-C-45662.

4.6.2.3.1 Functional.

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4.6.2.3.1.1 Power. The VS shall be tested in accordance with Appendix Test Method 1. Nonconformance to 3.6.2.1 shall constitute failure of this test.

4.6.2.3.1.2 Performance tests. The VS shall be tested in accordance with Appendix Test Method 2. Nonconformance to 3.6.2.2 shall constitute failure of this test.

4.6.2.3.1.3 Detailed circuit test. The VS shall be tested in accordance with Appendix Test Method 3. Nonconformance to 3.6.2.3 shall constitute failure of this test.

4.6.2.3.1.4 Tamper switch tolerance test. The VS shall be tested in accordance with Appendix Test Method 4. Nonconformance to 3.6.2.4 shall constitute failure of this test.

4.6.2.3.2 Detector circuit card assembly. The detector circuit card assembly shall be tested in accordance with Appendix Test Method 6. Nonconformance to 3.6.2.5 shall constitute failure of this test.

4.6.2.3.3 Processor circuit card assembly. The processor circuit card assembly shall be tested in accordance with Appendix Test Method 7. Nonconformance to 3.6.2.6 shall constitute failure of this test.

4.6.2.3.4 Environmental. The VS and major components (when furnished separately), shall be subjected to the following series of tests in accordance with MIL-STD-810. Prior to and following each test, conduct the operational test in accordance with Appendix Test Method 5.

NOTE: During the high and low temperature tests of 4.6.3.4.1 and 4.6.3.4.2, it is permissible to remove the assembly from the temperature chamber to conduct the tests of para. 2(i), 2(j), and 2(k) of Appendix Test Method 5.

4.6.2.3.4.1 Low temperature. Conduct the low temperature Test Method 502.1, Procedure I, storage temperature -30° F, for not less than 48 hours, equipment operating temperature -20° F. Major components (when furnished separately), shall be subjected to -20° F operational test. At the operating temperature perform the operating test in accordance with Appendix Test Method 5. The readings shall be within ±3dB of the recorded attenuator reading of Step (h) of Test Method 5, measured at room temperature (75° F ±5° F) prior to this environmental test.

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4.6.2.3.4.2 High temperature. Conduct the high temperature Test Method 501.1, Procedure II, except that the nonoperating temperature shall be +165° F; the operating temperature shall be +150° F. Major components (when furnished separately), shall be subjected to +150° F operational test. At the operating temperature perform the operating in accordance with Appendix Test Method 5. The readings shall be within ±3 dB of the recorded attenuator reading of Step (h) of Test Method 5, measured at room temperature (75° F ±5° F) prior to this environmental test.

4.6.2.3.4.3 Humidity test. Conduct Test Method 507.1, Procedure I with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Conduct the operating tests in accordance with Appendix Test Method 5, at completion of each cycle.
(b) During Steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
(c) During Step 4, the relative humidity shall be maintained at 92 percent or greater.
(d) Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. For the Tamper Test, the covers of the VS may be wiped of excess external moisture prior to the performance of this test. This test may be the last functional test performed prior to continuation of humidity cycling. The VS may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or unsuccessful return of the VS or major components (when furnished separately), to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test.

4.6.2.3.4.4 Shock.

(a) Basic design: Conduct Test Method 516.2, Procedure I, equipment not operating, sawtooth shock pulse, peak value 20 g's duration 11 ms.
(b) Bench handling: Conduct Test Method 516.2, Procedure V.

4.6.2.3.4.5 Vibration. Conduct Test Method 514.2, Procedure X, Curve AX, equipment not operating.

4.6.2.3.4.6 Failure criteria. Nonconformance to 3.5, or nonconformance to the requirements of Appendix Test Method 5 shall constitute failure of these tests.

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4.6.2.3.5 Electromagnetic susceptibility. The VS shall be subjected to the tests of MIL-STD-461 as follows.

4.6.2.3.5.1 Radiation susceptibility. Conduct Test Method RS03. Electric field requirements of RS03 shall be modified and limited as specified in 3.7.1. All leads (sensor, alarm and power) external to the signal processor enclosure shall be exposed to the electric field. The leads shall be 4 or more feet in length.

4.6.2.3.5.2 Conducted susceptibility. Conduct Test Methods CS01, CS02 and CS06 as modified in 3.7.2.

4.6.2.3.5.3 Failure criteria. The contractor shall furnish the contracting officer a report of the tests required by MIL-STD-461. Disapproval of the report shall constitute failure of these tests.

4.6.2.4 Reliability. Using an upper test MTBF (θ_0) of 10,000 hours and a lower test MTBF (θ_1) of 5,000 hours the VS shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-STD-781C, Test Plan VIIIC. Continue testing until either an "accept" or "reject" decision is reached. A reliability failure is defined as any malfunction which causes or may cause one or more of the following:

(a) Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
(b) Serious damage to the system.
(c) Serious personnel safety hazards.

4.6.2.4.1 Test procedure. The tests shall be conducted under ambient laboratory conditions, without temperature, humidity or vibration cycling. Equipment shall be energized at the primary power input specified in 3.6.2.1. Operation shall be continuous throughout the test. The 5 second test signal in 2(f) of Appendix Test Method 5 shall be applied to the VS not less than nine times during each 24 hour period. At the end of the reliability test the VS shall be subjected to the tests specified in 4.6.2.3.1.1 and 4.6.2.3.1.2. The tests shall commence after installation and a burn-in period of 72 hours as specified in 4.6.2.1. Parts or components of known limited life, such as batteries, shall have a life stipulated prior to test. Failures of these parts occurring after the end of the stipulated period are not relevant, but any dependent failures caused thereby are relevant. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause

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of the alarm identified if practicable. This accounting shall be maintained separately from the failure accounting. Redundant elements, such as indicator lamps, shall not be repaired nor counted as a failure until the VS has failed to perform its intended function. Preventive maintenance procedures specified for the equipment during normal operation shall be performed during the reliability test. No additional preventive maintenance is allowed during the test.

4.6.2.4.2 Failure criteria. Nonconformance to 3.8 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select VS's at any time during the contract production period and subject these VS's to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on VS's selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production VS's. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison VS to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison VS and shall be cause for refusal by the Government to continue acceptance of production VS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on VS's previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all VS's accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

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4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

111. Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
112. Boxed contents not cushioned to prevent damage as specified for Level A or Level B.
113. Strapping not as specified for Level A.
114. Quantities packed together exceed the size or weight limitation of the box for Level A or Level B.
115. Marking missing, illegible, incorrect, or incomplete for Level A or Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. The VS components shall be preserved by the applicable following procedure, depending upon procurement of complete VS's or procurement of components furnished separately.

5.1.1.1 Procurement of complete VS's. Each vibration detector and each signal processor shall be preserved in accordance with MIL-P-116, Method 11c. The two components shall then be placed together in a close-fitting box conforming to PPP-B-636, W6c or W5c, style optional. The components shall be cushioned within the box to prevent movement or damage.

5.1.1.2 Major components furnished separately. Major components furnished separately shall be individually preserved in accordance with MIL-P-116, Method IIc. Each component shall then be placed in a close-fitting box conforming to PPP-B-566, Variety 1, style, type, and class optional or PPP-B-676, Variety 1, style, type, and class optional. The components shall be cushioned to prevent movement or damage within the box.

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5.1.1.2.1 Consolidation. Major components of like description, preserved as specified in 5.1.1.2. shall be placed together in a close-fitting box conforming to PPP-B-636, W5c, style optional, in quantities not to exceed the weight or size limitation of the box.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for Level A, except Method IA-8 shall be used in place of Method IIc.

5.1.3 Commercial. Each complete VS and each major component furnished separately shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. The VS components, preserved as specified in 5.1, shall be packed by the applicable following procedure depending upon procurement of complete VS's or procurement of VS major components furnished separately.

5.2.1.1 Packing of complete VS's. Complete VS's, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional in quantities not to exceed the weight limitation of the box. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.1.2 Spare parts. VS major components furnished separately, preserved as specified in 5.1, shall be packed together as specified in 5.2.1.1. Only components of like description shall be packed together.

5.2.2 Level B. Complete VS's and major components furnished separately, preserved as specified in 5.1 shall be packed as specified in 5.2.1.1 or 5.2.1.2 as applicable, except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Complete VS's and major components furnished separately preserved as specified in 5.1, shall be in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.3.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188 and the special marking specified in 5.3.3.

5.3.3 Special marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

6. NOTES

6.1 Intended use. The vibration sensor is intended for use as a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When major components are to be furnished separately and number to be furnished (see 3.1.1).
(c) Time frame required for submission of preproduction model and number of VS's or major components (when furnished separately), required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of VS's to be furnished, when applicable (see 3.3).
(f) When Government-loaned property is not required (see 3.10).
(g) When preproduction-reliability tests can be deleted (see Table I).
(h) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(i) When other than Finish B strapping is required (see 5.2.1.1 and 5.2.1.2).

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6.3 Preproduction model. Any changes or deviations of production VS's and major components (when furnished separately), from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish VS's and major components (when furnished separately), conforming to this specification.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the VS is designed to alarm.

6.4.2 Tamper alarm. Alarm resulting from removal of the sensor assembly cover or pry-off of the sensor assembly from its mounting.

6.4.3 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, change of fit of any component or part, or any physical condition resulting in nonconformance to the provisions of this specification.

6.4.4 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP's indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of the repair parts.

6.5 Supplementary quality assurance provisions (SQAP) records. The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each VS.

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6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of VS's.

6.8 Government-loaned property. The contracting officer shall arrange to loan the property specified in 3.10, when applicable (see 6.2).

6.9 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodians:
Army - ME
Navy - EC
Air Force - 99

User activities:
Army - ER
Navy - SH, MC, CG

Preparing activity:
Army - ME

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APPENDIX

FOR

VIBRATION SIGNAL PROCESSOR

AND DETECTOR

10. SCOPE

10.1 Scope. The test methods contained within this appendix are to determine whether vibration sensors and major components furnished under this specification conform to the requirements set forth.

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TEST METHOD 1

POWER TEST

1. TEST APPARATUS:

   (a) Power supply, 20 ±2 Vdc, 100 mA.
   (b) Ammeter, 100 mA.
   (c) Digital voltmeter, dc.

2. TEST PROCEDURE:

   (a) Connect 20 detectors to TB2 of the processor.
   (b) Place the VS in operation by applying 20 Vdc through the ammeter to terminals TB1-7 and TB1-8 of the processor. The current shall not exceed 60 mA.
   (c) Reverse the polarity on terminals TB1-7 and TB1-8 for 10 seconds. The current shall not exceed 1 mA.
   (d) Reconnect as in Step (b). The current shall not exceed 60 mA.
   (e) Measure the voltage at TP4 with a voltmeter. The voltage shall be 12 ±0.6 Vdc.
   (f) Vary the voltage at TB1-7 and TB1-8 from 18 to 22 volts. The voltage at TP4 shall not vary more than ±100 mV.

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TEST METHOD 2

PERFORMANCE TESTS

1. TEST APPARATUS:

(a) Power supply, 20 to 30 Vdc, 100 mA (2 required).
(b) Signal generator, 5 to 20 kHz.
(c) Broadband amplifier.
(d) Broadband acoustic driver, Altec Model 802-8D and enclosure.
(e) Accelerometer, calibrated at 5, 10, 12, 15 and 20 kHz. Endevco Model 2217E or equivalent.
(f) Oscilloscope.
(g) Voltmeter, ac, Hewlett-Packard 400E or equivalent.

2. TEST PROCEDURE:

(a) Alarm criteria.

(1) Place 19 of 20 detectors in a vibration-free area. One test detector shall be excited by the vibration test signals.
(2) Place operating test detector on driver enclosure, ensuring detector is aligned with output post of driver.
(3) Connect all 20 detectors to TB2 of the processor by 500 feet of cable.
(4) Set the processor sensitivity to maximum (5) and the test detector gain to 10 dB below maximum (7.5).
(5) The vibration test signal is applied by connecting the signal generator, through the broadband amplifier, to the acoustic driver. Energy from the acoustic driver, which is located within 1/2 inch from the detector vibrator transducer, excites the transducer. Adjust the broadband amplifier's output for 0.03g acceleration on the transducer plate at the desired frequency using the calibration data for the test setup.
(6) Apply 30 Vdc, through 2.7k ohms, to terminals TB1-1 and TB1-2 (processor), and connect oscilloscope across these terminals.
(7) Place the VS in operation by applying 20 Vdc to terminals TB1-7 and TB1-8 (processor).

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(8) Apply a 5-second burst of vibration test signal at 12 kHz and 0.03g. The VS shall alarm.
(9) Apply a 100 ms burst of test signal (12 kHz and 0.03g) every 4 seconds. The VS shall alarm on the fourth burst. As indicated on the oscilloscope, the alarm duration shall be from 0.5 to 1.5 seconds.

(b) Bandpass characteristics.

(1) With test signal applied as in 2(a)(4) and the VS in operation, connect the ac voltmeter to TP1. Set processor sensitivity to 3.
(2) Adjust the 12 kHz test signal amplitude to obtain 0.5V root mean square (rms) at TP1.
(3) Vary the signal generator frequency to 10 kHz and 15 kHz. The response at TP1 shall not differ by more than ±3 dB from the average in-band response.
(4) Keeping the test signal level constant, as in Step (b), set the test signal frequency to 5 kHz and then to 20 kHz. For both frequencies, the response at TP1 shall be at least 30 dB below the average in-band response.

(c) Sensitivity control range.

(1) Repeat Step (b)(1), but set processor sensitivity to 5.
(2) Adjust the 12 kHz test signal amplitude to obtain 1.0V rms at TP1.
(3) Set processor sensitivity to minimum (1). The signal level at TP1 shall decrease at least 24 dB.

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TEST METHOD 3

DETAILED CIRCUIT TESTS

1. TEST APPARATUS:

(a) Power supply, 20 to 30 Vdc, 100 mA (2 required).
(b) Voltmeter, ac.
(c) Signal generator.
(d) Step attenuator.
(e) Oscilloscope.
(f) Voltmeter, dc.
(g) Ammeter, dc, (30 mA).

2. TEST PROCEDURE:

(a) Detector gain control.

(1) With processor and detector interconnected, connect test signal from signal generator, through step attenuator and 0.1 microfarad (uF) capacitor, to "RED" input pin in detector.
(2) Set processor sensitivity to 3 and detector gain to maximum (10).
(3) Place VS in operation by applying 20 Vdc to processor terminals TB1-7 and TB1-8.
(4) Set test signal to 12 kHz and adjust input level to obtain 1 V rms at TP1.
(5) Turn detector gain control to minimum (1). The signal level at TP1 shall decrease by at least 20 dB.

(b) Alarm duration.

(1) With equipment connected as in Step (a)(1), apply 30 Vdc, through 2.7k ohms, to processor terminals TB1-1 and TB1-2, and connect oscilloscope across these terminals.
(2) With processor sensitivity set to 3 and detector gain set to 7.5, and VS in operation, adjust 12 kHz test signal amplitude to obtain 1 V rms at TP1. Remove test signal.

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(3) Apply four 100 ms bursts of test signal within a 15-second period. The oscilloscope shall indicate an alarm on the fourth burst. Observe the time duration of the alarm. The alarm duration shall be from 0.5 to 1.5 seconds.

(c) Alarm output.

(1) With equipment connected as in Step (a)(1), apply 30 Vdc through 2.7k ohms and the ammeter, to processor terminals TB1-1 and TB1-2, and connect dc voltmeter across these terminals.
(2) Repeat Step (b)(2).
(3) In no-alarm state, ammeter shall indicate at least 10 mA, demonstrating 10 mA capability and resistance less than 300 ohms.
(4) Momentarily interrupt 20 Vdc power, causing VS to alarm. Voltmeter shall read 30 Vdc, demonstrating 30 Vdc hold and resistance greater than 100k ohms.

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TEST METHOD 4

TAMPER SWITCH TOLERANCE TEST

1. TEST APPARATUS:

   (a) Ohmmeter.
   (b) Test covers, with holes for meter lead access, for processor and detector.

2. TEST PROCEDURE:

   (a) Connect an ohmmeter between processor tamper terminals TB1-5 and TB1-6.
   (b) Close the test cover of the processor enclosure. The resistance across the tamper terminals (TB1-5 and TB1-6) shall be less than 100 ohms.
   (c) The resistance between TB1-5 and TB1-6 shall increase to greater than 100k ohms when the cover is raised more than 0.062 inch but not more than 0.250 inch (flanges engaged).
   (d) The resistance between TB1-5 and TB1-6 shall be greater than 100k when the cover is opened.
   (e) Pull the tamper switch fully out and lock. The resistance between TB1-5 and TB1-6 shall decrease to less than 100 ohms.
   (f) Repeat Step (b) and (c).

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TEST METHOD 5

OPERATIONAL TEST

1. TEST APPARATUS:

(a) Power supply, 20 ±2 Vdc, 100 mA.
(b) Power supply, 30 ±2 Vdc, 20 mA.
(c) Oscilloscope, dc coupled, 10 volts per centimeter (V/cm), 200 milliseconds per centimeter (ms/cm).
(d) Signal generator, to 20 kHz.
(e) Step attenuator.
(f) Voltmeter, ac.
(g) Broadband amplifier.
(h) Broadband acoustic driver, Altec Model 802-8D.
(i) Accelerometer, calibrated at 5, 10, 12, 15 and 20 kHz. Endevco Model 2217E or equivalent.

2. TEST PROCEDURE:

(a) With processor and detector interconnected, connect test signal from signal generator, through step attenuator and 0.1 uF capacitor, to "RED" input pin in detector.
(b) Set processor sensitivity to 3, and detector gain to 7.5.
(c) Apply 30 Vdc, through 2.7 ohms, to processor terminals TB1-1 and TB1-2 and connect oscilloscope across these terminals.
(d) Place VS in operation by applying 20 Vdc to processor terminals TB1-7 and TB1-8.
(e) Set test signal to 12 kHz, and adjust input level to obtain 1.2 V rms at TP1 and record the step attenuator reading.
(f) Observe alarm output (on oscilloscope) which should be cycling. Remove test signal and momentarily short TP2 to ground. Then apply a 5-second test signal at 12 kHz. The VS shall alarm.
(g) Momentarily short TP2 to ground. Apply 100 ms test signals at 4- to 5-second intervals. The VS shall alarm on the fourth test signal. The alarm duration (as observed on the oscilloscope) shall be from 0.5 to 1.5 seconds.
(h) Vary the test signal frequency from 10 to 15 kHz. The signal amplitude at TP1 shall be 1.2 V rms ±3 dB.

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(i) Apply a 5-second burst of vibration test signal at 12 kHz and 0.03g. The VS shall alarm. The vibration test signal is applied by connecting the signal generator, through the broadband amplifier, to the acoustic driver. Energy from the acoustic driver, which is located within 0.50 inch from the detector vibration transducer, excites the transducer. Adjust the broadband amplifier's gain for 0.03g acceleration on the transducer plate at the desired frequency using the calibration data for the test setup.

(j) Apply a 100 ms burst of test signal (12 kHz and 0.03g) every 4 seconds. The VS shall alarm on the fourth burst. As indicated on the oscilloscope, the alarm duration shall be from 0.5 to 1.5 seconds.

(k) Repeat (i) and (j) at 10 kHz and 15 kHz.

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TEST METHOD 6

CIRCUIT CARD ASSEMBLY, DETECTOR

(13220E3063)

1. TEST APPARATUS:

(a) Power supply, dc, Hewlett-Packard 6101A or equivalent.
(b) Signal generator, Wavetek Model 110 or equivalent.
(c) Oscilloscope, Tektronix 465 or equivalent.
(d) Multimeter, Simpson 260 or equivalent.
(e) Test fixture (see Figure 1).

2. TEST PROCEDURE (ROOM TEMPERATURE):

(a) Vibration detector circuit test.

(1) Arrange the equipment as shown in Figure 1. Make certain that the power supply is set to zero output before turning on. Raise the power supply output voltage to 5.7 volts.
(2) Adjust the signal generator to a frequency of 12 kHz. Connect the oscilloscope to the "RED" terminal of the circuit board. Adjust the generator to a signal level of 100 mV P-P. Set the GAIN knob on the circuit board assembly to position 10 (max gain). Connect the oscilloscope to terminal TB1-1. The output at this point shall be 30 mV P-P $\pm$3 dB. Set the gain knob to position 1. Maintain the input signal level at 100 mV P-P. The output shall be 2 mV P-P $\pm$3 dB. The output signal amplitude at GAIN knob setting 1 must be at least 20 dB less than the level at position 10.

(b) Vibration detector tamper switch test.

(1) Connect an ohmmeter between terminals TB1-5 and TB1-6. The ohmmeter shall indicate greater than 100k ohms.
(2) Manually depress the tamper switch push button. The ohmmeter shall indicate less than 100 ohms.
(3) Pull the tamper switch push button out to the service position. The ohmmeter shall indicate less than 100 ohms.

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3. TEST PROCEDURE (-20° F):

Procedures and specifications are identical to paragraph 2 except circuit assembly in environmental chamber shall be "soaked" at -20° F for at least 1 hour before testing begins.

4. TEST PROCEDURE (+150° F):

Procedures and specifications are identical to paragraph 2 except circuit assembly in environmental chamber shall be "soaked" at +150° F for at least 1 hour before testing begins.

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TEST METHOD 7

CIRCUIT CARD ASSEMBLY, PROCESSOR

(13220E3013)

1. TEST APPARATUS:

(a) Power supply, dc, Hewlett-Packard 6101A or equivalent.
(b) Signal generator, Wavetek Model 110 or equivalent.
(c) Oscilloscope, Tektronix 465 or equivalent.
(d) Step attenuator, Hewlett-Packard 355D or equivalent.
(e) Multimeter, Simpson 260 or equivalent.
(f) Stop watch.
(g) Test fixture (see Figure 2).

2. TEST PROCEDURE (ROOM TEMPERATURE):

(a) Bandpass characteristics.

(1) Arrange the equipment as in Figure 2. Make certain that the power supply is set to zero output before turning on. Raise the power supply output voltage to 20 Vdc. Place the voltmeter on TP4. The voltage reading shall be 12 ±1 Vdc.
(2) Place the oscilloscope probe on Test Point 1. Set the sensitivity control knob on the processor circuit card to 3. Place the continuous/momentary switch in the continuous position. Adjust the attenuator for a total attenuation of 90 dB. Set the signal generator to 12 kHz. Adjust the amplitude of the signal generator output for a value of 0.5 V rms (1.43 V P-P) at TP1. Measure the amplitude of the generator output. The measured value shall be 1 Vac ±6 dB.
(3) Vary the signal generator frequency between 10 and 15 kHz. Note the highest amplitude produced as measured at TP1. The values of signal amplitude at 10 and 15 kHz shall not be more than 6 dB below the maximum observed value. Set the signal generator frequency at 5 kHz. The signal amplitude at TP1 shall be at least 30 dB below the peak

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amplitude which occurred between 10 and 15 kHz. Set the signal generator frequency at 20 kHz. The signal amplitude at TP1 shall be at least 30 dB below the peak amplitude which occurred between 10 and 15 kHz.

(4) Turn the sensitivity knob on the processor circuit board to position 5. Adjust the attenuator for a total attenuation of 110 dB. Set the signal generator to 12 kHz. Adjust the amplitude of the signal generator output for a value of 0.5 V rms (1.43 V P-P) at TP1. Measure the amplitude of the generator output. The measured value shall be 2.5 Vac ±6 dB.

(5) Repeat Step (3).

(b) Sensitivity control range.

(1) Confirm that the sensitivity knob on the processor circuit board is on position 5. Adjust the attenuator and/or the signal generator output level to produce a level of 1 V rms at TP1. (Signal generator set at 12 kHz.) Turn the sensitivity knob on the processor circuit board to position 1. The signal level at TP1 shall have decreased by at least 24 dB.

(c) Alarm duration.

(1) Multivibrator test.

a. Arrange the equipment as shown in Figure 2. Turn the sensitivity knob on the processor circuit board to position 3. Adjust the attenuator and/or the signal generator output amplitude to produce a level of 1 V rms at TP1 (signal generator frequency set at 12 kHz). Place the momentary/continuous signal control switch in the center-off position. Connect the oscilloscope probe to test point MV (multivibrator).

b. Actuate the signal control switch to the momentary on position. An 11 V ±10 percent pulse of approximately 1.5 seconds duration shall be produced each time the switch is actuated.

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(2) Continuous alarm input test.

a. Connect the oscilloscope probe to test point INT (integrator).
b. Hold the signal control switch in the momentary on position and simultaneously start the stop watch. The oscilloscope shall indicate that the integrator has reached the 3 volt threshold level and resets to zero in 4.5 ±0.5 seconds. Release to the signal control switch to the center off position.

(3) Random alarm input test.

a. Confirm that test point INT signal level is resting at zero volts ±0.1 volt.
b. Actuate the signal control switch to the momentary on position. Do not hold the switch in the on position for more than 1 second. Wait 4 seconds from the first initiation of the switch and repeat. Wait an additional 4 seconds and actuate the switch a third time. Wait another 4 seconds and actuate the switch a fourth time. The oscilloscope shall indicate that the integrator has reached the 3 volt threshold level and resets to zero upon the generation of the fourth pulse.

(d) Alarm output test.

(1) Connect a clip lead from TB1-7 (20 V source) to TB1-1 (alarm relay terminal). Connect the oscilloscope test probe to TB1-2. Place the signal control switch in the continuously on position.
(2) The oscilloscope shall indicate a 20 volt pulse with a duration from 0.5 to 1.5 seconds occurring at intervals of approximately 4.5 seconds.

(e) Vibration processor tamper switch test.

(1) Connect an ohmmeter between terminals TB1-5 and TB1-6. The ohmmeter shall indicate greater than 100k ohms.

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(2) Manually depress the tamper switch push button. The ohmmeter shall indicate less than 100 ohms.
(3) Pull the tamper switch push button out to the service position. The ohmmeter shall indicate less than 100 ohms.

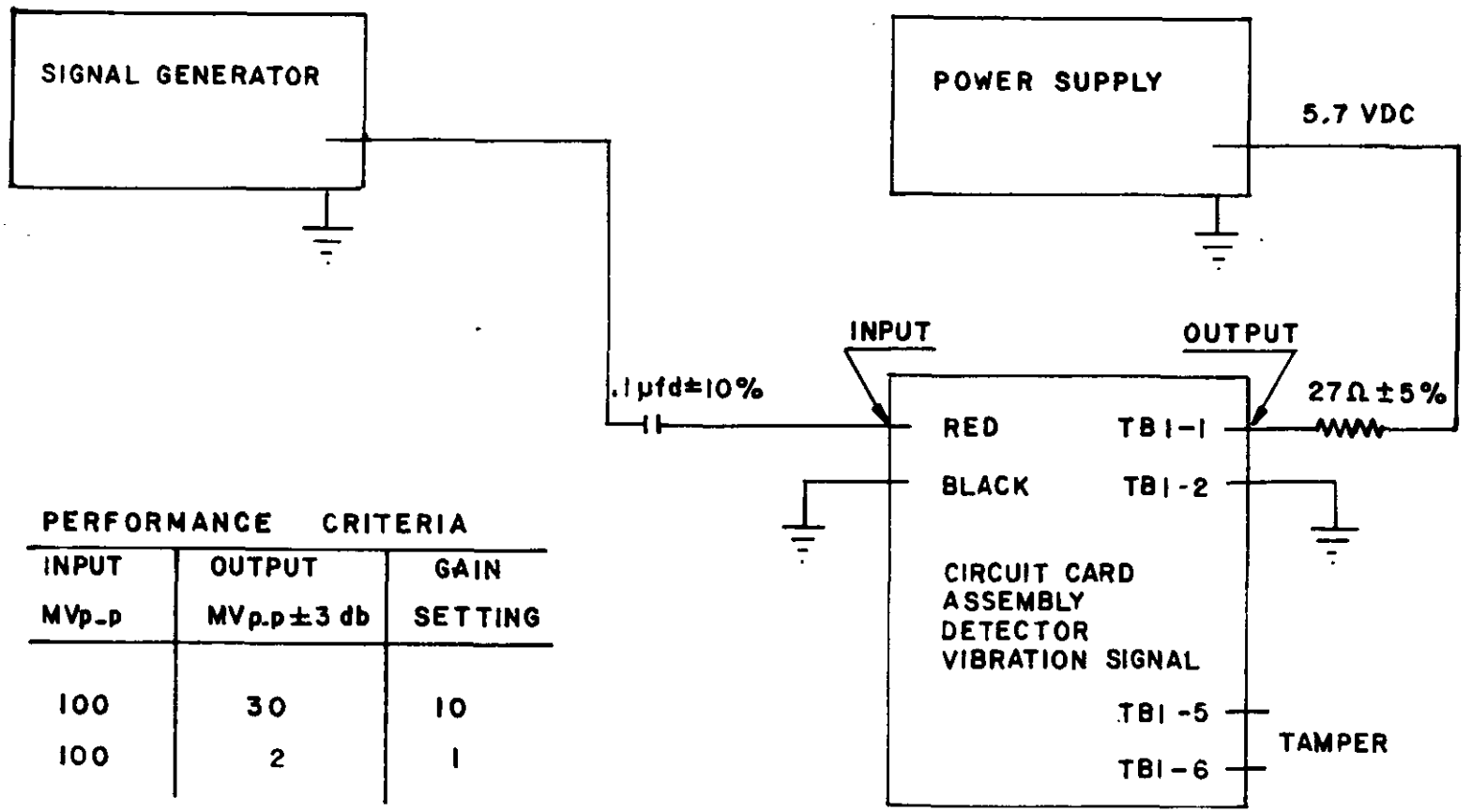
3. TEST PROCEDURE (-20° F):

Procedures are identical to paragraph 2 except circuit assembly in environmental chamber shall be 'soaked' at -20° F for at least one hour before testing begins. Specifications are identical except for voltage requirement on TP-1 stated in paragraph (c)(1)(a). At -20° F voltage level at TP-1 shall read 1.2 $\pm$0.2 volts rms for actuation of multivibrator.

4. TEST PROCEDURE (+150° F):

Procedures are identical to paragraph 2 except circuit assembly in environmental chamber shall be 'soaked' at +150° F for at least 1 hour before testing begins.

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PERFORMANCE CRITERIA

INPUT MVp-p	OUTPUT MVp-p ±3 db	GAIN SETTING
100	30	10
100	2	1

FIGURE 1. Test fixture for card assembly, detector (13220E3063). X-3932A

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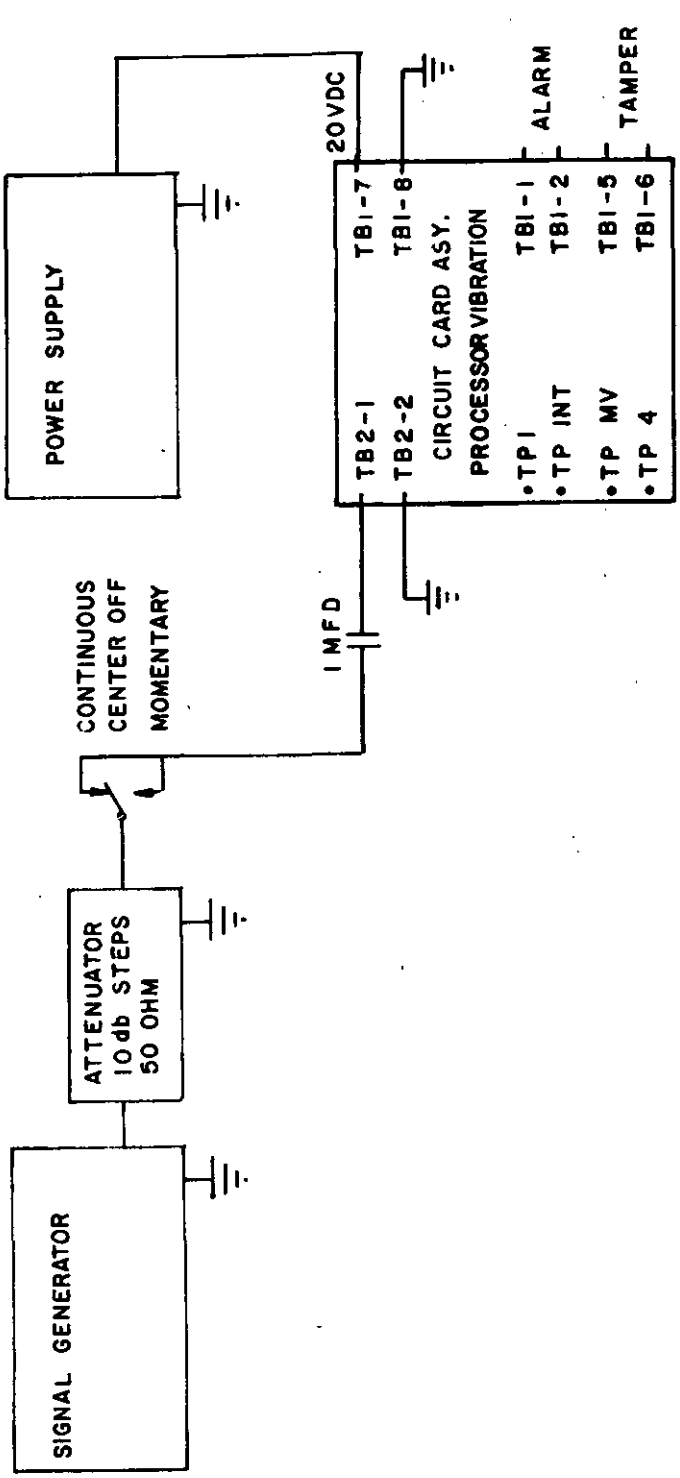


FIGURE 2. Test fixture for circuit card assembly processor (1322OE3013).

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